

Black holes and Relativists

$\sim 20,000$ stars

$\hookrightarrow$ most become black holes

Can learn how many black holes are near us

$\hookrightarrow$ closest should be about ~ 10 pc

Lots of black holes are proposed $\sim$ transients

$\hookrightarrow$ dynamical confirmation helps discovery

Mass gap: $2 \sim 5.5 M_{\odot}$ Why?

$100 + M_{\odot}$: How did they form?

Grain:

- Very fine precision

- fits keplerian orbit

$\hookrightarrow$ no astrometric measurements $\rightarrow$ well constrained parameters

Expands literature

~ 5000 WD candidates; 90% confirmed

~ 50 neutron stars; 30 confirmed

~ 5 black holes; 2 confirmed

BH1

How to form?

$\hookrightarrow$ too tight of binary for star to begin orbit

- strong winds?

- triplet?

- captured later?

BH: three discovered

- Can get progenitor metallicity

What if two blackholes?

- ↳ If two will deviate from keplerian orbit
- ↳ No evidence yet → will keep looking
- ↳ Only feasible way to find binary

Based on binaries found, likely that much more common than X-ray binaries

Mass gap persists even with new data
Metallicity. It's complicated

Blackhole kicks:

- Most have large kicks

DR4: It's coming

- More stars, planets, and blackholes!

Roman:

- should discover isolated blackholes
- Roman data can constrain orbits

Q1) If mass gap persists even with Gaia data release 4, what might that say about why the mass gap is there?

Q2) How would an estimate of isolated black holes change current theories on black hole evolution?